AQUAFAB INDUSTRIES

Running a Temperature Program

Tidal cycles, ramps and recorded field traces.

For Aquafab Workbench 0.1.0

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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1 Running a temperature program

A temperature program makes the chamber follow a curve: a tidal cycle, a ramp, a series of holds, or a temperature log recorded in the field. It can run on its own, recording temperature and flow, or alongside respirometry.

> Screenshots are from a simulated machine. Numbers in them are not measurements.

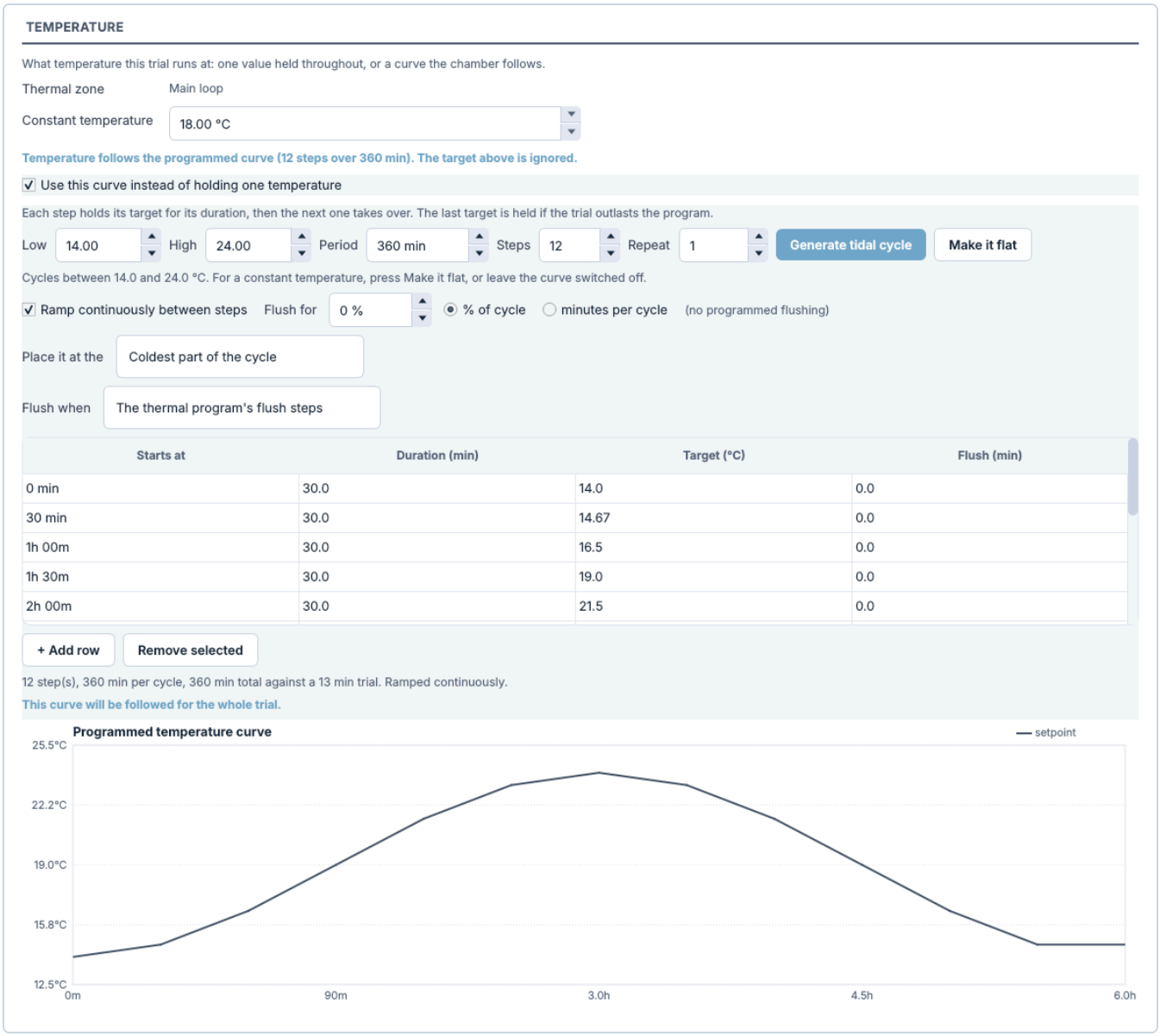
1.1 Before you start

Run the **Thermal Step Test** once on the machine (*Commissioning a New Machine*, step 4). It measures how fast the chamber can heat and cool, and the program page uses that to warn when a curve asks for more.

1.2 Build the curve

Open the machine, then **Experiments → Respirometry → Set up**.

1. In **What are you measuring?**, choose **Temperature program only**. (Or **Respirometry** to measure oxygen during the program.)
2. In **This experiment**, give it a **Name** and **Treatment**.
3. Go to the **Temperature** card and tick **Use this curve instead of holding one temperature**.



The Temperature card with a tidal cycle generated.

1.2.1 Generate a tidal cycle

Field	What to enter
Low, **High**	The extremes of the cycle.
Period (min)	One full cycle, for example 720 for a 12-hour tide.
Steps	How many holds the curve is cut into. More is smoother.
Repeat	How many cycles to run back to back.

Press **Generate tidal cycle**. Tick **Ramp continuously between steps** for a smooth curve rather than a staircase. **Make it flat** gives a long hold at one temperature.

1.2.2 Edit by hand

The program table lists every step: **Starts at**, **Duration (min)**, **Target (°C)** and **Flush (min)**. Edit any cell; **+ Add row** and **Remove selected** change the steps. The preview chart redraws as you edit.

1.2.3 Flushing during a long program

A long program in a closed chamber drives the animal hypoxic. Set:

- **Flush for**: as **% of cycle** or **minutes per cycle**;
- **Place it at the**: *coldest part* (the tide returning), *warmest part*, *start*, *end*, or *spread evenly*;
- **Flush when**: the program's flush steps, when oxygen drops below a setpoint (needs a probe), or whichever comes first.

Flushing is suspended during respirometry measurement windows.

1.3 Check it against the machine

Under the table:

- **A rate warning**, "*This curve is steeper than measured… The water will lag behind it*", means a step asks for faster heating or cooling than the step test measured. Lengthen the period or narrow the range.
- **A reservoir note**, on machines with a chilled reservoir, says how cold to chill it before starting.

Set **Water speed** if the flow should be held or stepped through the program. Press **Save experiment** to keep it.

1.4 Run

1. Switch to **Live** and press **Prepare system**. It brings the chamber to the program's first temperature and chills the reservoir. Wait until ready.
2. Press **Start Trial**.
3. The **Programmed curve** chart shows the curve, the measured temperature over it, and how hard the heater, exchanger pump and chiller are working.
4. When it finishes, press **Save Trial**. **It is not saved until you do.**

1.5 Follow a recorded field trace

To make the chamber follow a temperature log from a tide pool or a logger:

1. **This experiment → Import readings from a file…** and choose the CSV. In the dialog, choose the **Time column** and its unit, tick the readings to import and say what each is measured in. The import is kept as a trial, marked *imported* everywhere. (Needs a license.)
2. In **Trial History**, **Inspect** the imported trial and press **Use as a temperature program…**.

3. Choose the **Temperature series**, how many **Steps** the trace becomes, and how many times to **Repeat** it. Press **Use as the temperature program**.
4. The curve appears on the Temperature card. Check it, then prepare and run as above.